

# Nelson–Siegel–Svensson Estimation of Greek, Indian, and Italian Sovereign Discount Curves

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## Abstract

This paper presents a reproducible comparison of Greek, Indian, and Italian sovereign discount curves. Official benchmark yields are transformed according to an explicit modelling convention, fitted separately with the Nelson–Siegel–Svensson model, and converted into model-implied discount factors. The empirical tables and figures are generated automatically by the repository pipeline.

## 1 Discount Factors

**Definition 1.1** (Discount factor). *For maturity  $m$ , the discount factor  $D(m)$  is the present value of one unit payable at maturity under a specified pricing convention.*

Under continuous compounding,

$$D(m) = \exp[-my(m)], \quad (1)$$

where  $y(m)$  is the continuously compounded spot yield.

Observed sovereign benchmark yields-to-maturity are not themselves directly observed zero-coupon spot rates. Accordingly, the repository records its transformation convention explicitly and interprets all resulting discount factors as model-implied quantities.

## 2 Nelson–Siegel–Svensson Specification

Let  $\theta = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$  with  $\tau_1, \tau_2 > 0$ . The NSS spot curve is

$$y(m; \theta) = \beta_0 + \beta_1 \frac{1 - e^{-m/\tau_1}}{m/\tau_1} \quad (2)$$

$$+ \beta_2 \left( \frac{1 - e^{-m/\tau_1}}{m/\tau_1} - e^{-m/\tau_1} \right) \quad (3)$$

$$+ \beta_3 \left( \frac{1 - e^{-m/\tau_2}}{m/\tau_2} - e^{-m/\tau_2} \right). \quad (4)$$

For each country  $c \in \{\text{GR}, \text{IN}, \text{IT}\}$ , the model-implied discount curve is

$$\hat{D}_c(m) = \exp[-my(m; \hat{\theta}_c)]. \quad (5)$$

The three-way discount-factor comparisons are

$$\Delta D_{GR,IT}(m) = \hat{D}_{GR}(m) - \hat{D}_{IT}(m), \quad (6)$$

$$\Delta D_{GR,IN}(m) = \hat{D}_{GR}(m) - \hat{D}_{IN}(m), \quad (7)$$

and

$$\Delta D_{IT,IN}(m) = \hat{D}_{IT}(m) - \hat{D}_{IN}(m). \quad (8)$$

### 3 Reproducible Empirical Pipeline

The workflow executes the sequence

raw data  $\rightarrow$  estimation  $\rightarrow$  plot generation  $\rightarrow$  tests  $\rightarrow$  generated results  $\rightarrow$  PDF  
compilation.

The empirical material below is generated automatically from the current repository outputs.

### 4 Reproducible June 2025 Greece–India–Italy Results

This section is generated automatically by the repository workflow from the empirical output files. The underlying inputs are benchmark yields transformed according to the repository’s declared modeling convention; the resulting curves are therefore NSS model-implied discount factors rather than directly observed zero-coupon discount factors.

Table 1: NSS estimation diagnostics generated by the empirical pipeline.

| Country | $\beta_0$  | $\beta_1$   | $\beta_2$   | $\beta_3$   | $\tau_1$ | $\tau_2$ | RMSE       |
|---------|------------|-------------|-------------|-------------|----------|----------|------------|
| Greece  | 0.35732110 | -0.34046812 | -0.14809965 | -0.81407869 | 5.5000   | 25.5000  | 0.00004435 |
| Italy   | 0.04604401 | -0.02165477 | -0.04818816 | 0.00020321  | 1.5000   | 8.0000   | 0.00000000 |
| India   | 0.07167323 | -0.03376065 | 0.02784213  | 0.04172842  | 9.5000   | 1.5000   | 0.00000264 |

### 5 Reproducibility and Interpretation

The pipeline succeeds only when empirical estimation, visualization, and the complete unit and regression suite succeed before paper publication. The resulting PDF therefore remains conditional on the tracked data, the declared transformation assumption, and the NSS implementation tested by continuous integration.

Table 2: NSS model-implied three-way discount-factor comparison.

| Maturity | Greece     | India      | Italy      | $D_{GR} - D_{IN}$ | $D_{GR} - D_{IT}$ | $D_{IT} - D_{IN}$ |
|----------|------------|------------|------------|-------------------|-------------------|-------------------|
| 0.5      | 0.99121119 | 0.97771237 | 0.98946411 | 0.01349882        | 0.00174709        | 0.01175173        |
| 1.0      | 0.98164827 | 0.95120315 | 0.98037536 | 0.03044512        | 0.00127292        | 0.02917221        |
| 2.0      | 0.96006845 | 0.89435385 | 0.96041786 | 0.06571460        | -0.00034941       | 0.06606401        |
| 3.0      | 0.93531750 | 0.83856300 | 0.93500818 | 0.09675449        | 0.00030932        | 0.09644518        |
| 5.0      | 0.87818206 | 0.73636356 | 0.87109869 | 0.14181850        | 0.00708337        | 0.13473513        |
| 7.0      | 0.81520694 | 0.64508462 | 0.80087065 | 0.17012232        | 0.01433629        | 0.15578603        |
| 10.0     | 0.72014715 | 0.52469503 | 0.69977250 | 0.19545213        | 0.02037466        | 0.17507747        |
| 15.0     | 0.58085573 | 0.36534439 | 0.55608047 | 0.21551134        | 0.02477526        | 0.19073608        |
| 20.0     | 0.46969468 | 0.25186856 | 0.44163295 | 0.21782612        | 0.02806173        | 0.18976439        |
| 30.0     | 0.29044017 | 0.11961911 | 0.27859393 | 0.17082106        | 0.01184624        | 0.15897482        |

## References

- [1] Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves. *Journal of Business*, 60(4), 473–489.
- [2] Svensson, L. E. O. (1994). Estimating and interpreting forward interest rates: Sweden 1992–1994. *NBER Working Paper*.

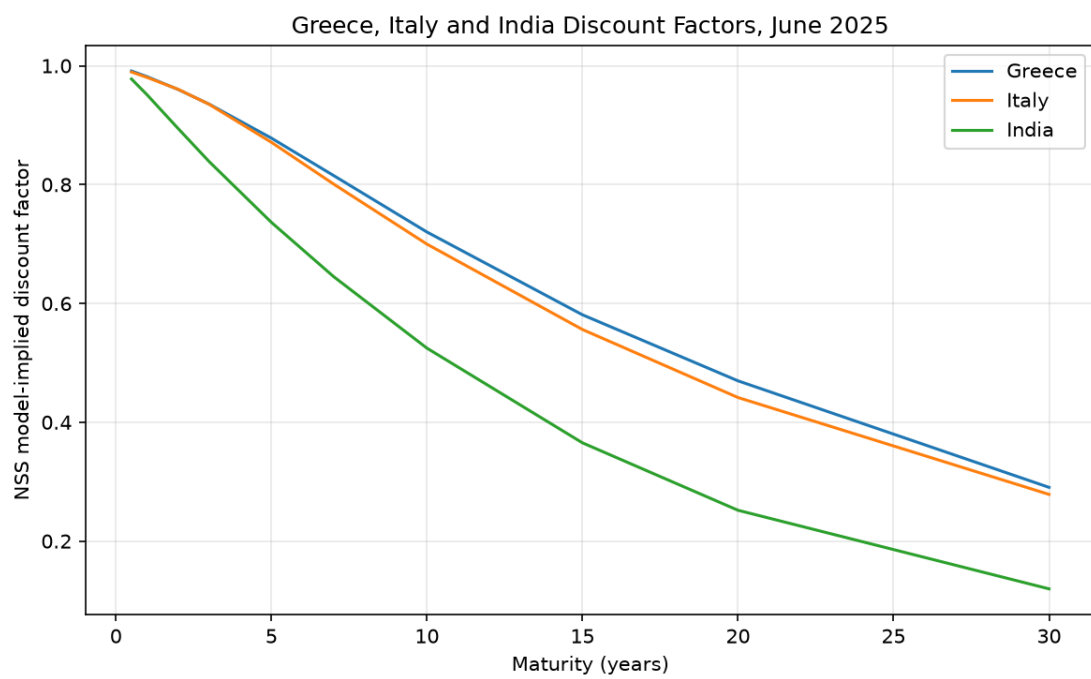


Figure 1: Automatically generated NSS model-implied Greece–India–Italy discount-factor comparison.